Data Visualisation and Visual Analytics

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Executive Summary

This report presents an analysis of a high-dimensional dataset related to Australian Open tennis championship match records. The dataset, covering the period from 1905 to 2023, includes attributes such as match year, gender, player performance (champion and runner-up), scores, win/loss counts, and win rates. Using data visualization techniques like Tree Maps, Parallel Coordinate Charts, and Geographic Maps, we explored trends in player performance, country dominance, and championship outcomes.

Key findings from the analysis include the dominance of Australian and American players in both men's and women's divisions, with Australia having won the most championships overall. Novak Djokovic, Margaret Court, and Serena Williams emerged as the top performers based on win rates and championship victories. The analysis also revealed a correlation between early set wins and overall match outcomes, with longer matches generally resulting in lower win rates.

By leveraging advanced visualizations, this report highlights the strengths and weaknesses of various players, countries, and match conditions, offering a comprehensive view of the factors that have shaped the Australian Open championships over the years. The findings provide valuable insights for understanding the dynamics of player performance and the influence of geographical factors in major tennis tournaments.

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Data Exploration:

Attribute_Name	Type	Description
Year	4 digit format in YYYY	Quantitative (Interval) - Year of The Tournamen
Gender	Categorical (Nominal)	Gender of the tournament (Men's/Women's)
Champion	Text	Name of the player who won the tournament
Champion Nationality	Text	Nationality of the player who won the tournament
Champion Country	Text	Country of the player who won the tournament
Score	Text	Scores in the format of sets won and lost
1st-won	Numeric (Discrete)	Number of games won in the first set by the winner
1st-loss	Numeric (Discrete)	Number of games lost in the first set by the winner
2nd-won	Numeric (Discrete)	Number of games won in the second set by the winner
2nd-loss	Numeric (Discrete)	Number of games lost in the second set by the winner
3rd-won	Numeric (Discrete)	Number of games won in the third set by the winner
3rd-loss	Numeric (Discrete)	Number of games lost in the third set by the winner
4th-won	Numeric (Discrete)	Number of games won in the fourth set by the winner
4th-loss	Numeric (Discrete)	) Number of games lost in the fourth set by the winner
5th-won	Numeric (Discrete)	Number of games won in the fifth set by the winner
5th-loss	Numeric (Discrete)	Number of games lost in the fifth set by the winner
Runner-up	Text	Name of the player who finished as runner-up
Runner-up Nationality	Text	Nationality of the player who finished as runner up
Runner-up Country	Text	Country of the player who finished as runner-up

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Understanding Dataset

Assignment Data

The dataset for this assignment comes from the Australian Open Championship, one of the most prestigious tennis tournaments in the world. It includes data spanning 118 years, from 1905 to 2023, for both men's and women's championship matches. The dataset is composed of 206 records and 22 attributes, covering various aspects of the matches, such as player details, match outcomes, and scores.

Key attributes include:

Year: The year the match took place.

Gender: Gender of the participants (Men's/Women's).

Champion Name: Name of the player who won the tournament.

Champion Nationality: Nationality of the winning player.

Champion Country: The country represented by the winning player.

Champion Seed: The player's seeding in the tournament.

Mins: Match duration in minutes.

Score: Final match score.

Set Wins/Losses: Number of games won or lost in each set by the champion and runner-up.

Runner-up Name: Name of the player who finished as the runner-up.

Runner-up Nationality/Country: Nationality and country represented by the runner-up.

Runner-up Seed: Seeding of the runner-up.

Additionally, three attributes were introduced for deeper analysis:

Win: Total wins in the championship game.

Loss: Total losses in the championship game.

Win-rate: The percentage of wins.

Data Format

The dataset includes a mix of data types:

Categorical Data: Attributes like "Gender," "Champion Name," "Nationality," and "Runner-up Name" consist of categorical information, which classifies players and nationalities.

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Interval Data: The "Year" attribute is interval data, representing the time span with meaningful differences between values.

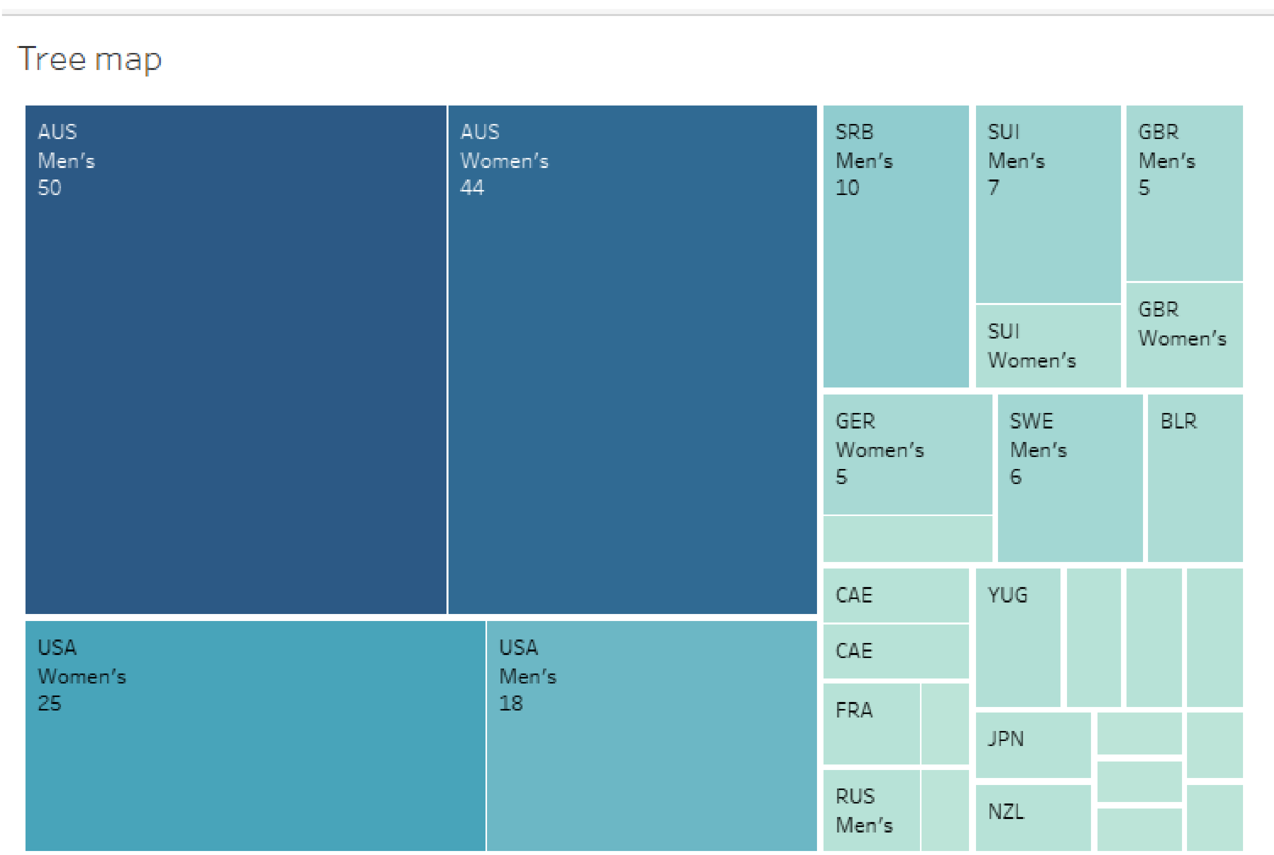
Ratio Data: Attributes like "Win," "Loss," and "Win-rate" are ratio data, where calculations like averages and proportions can be performed. These metrics are crucial for analyzing player performance over time.

Data Cleaning

Before visualizing the dataset, data cleaning was performed. Missing values in some attributes (e.g., Champion Seed, Match Duration, and Runner-up Seed) before 1945 were removed due to incomplete recording. All null values in the set win/loss columns were replaced with zeros, ensuring consistent data for analysis.In addition, the "Score" attribute was cleaned and transformed from text into separate numeric columns to accurately reflect game outcomes, further enhancingthe dataset for precise visual analysis.

Visualisation Techniques:

.1 Tree Map



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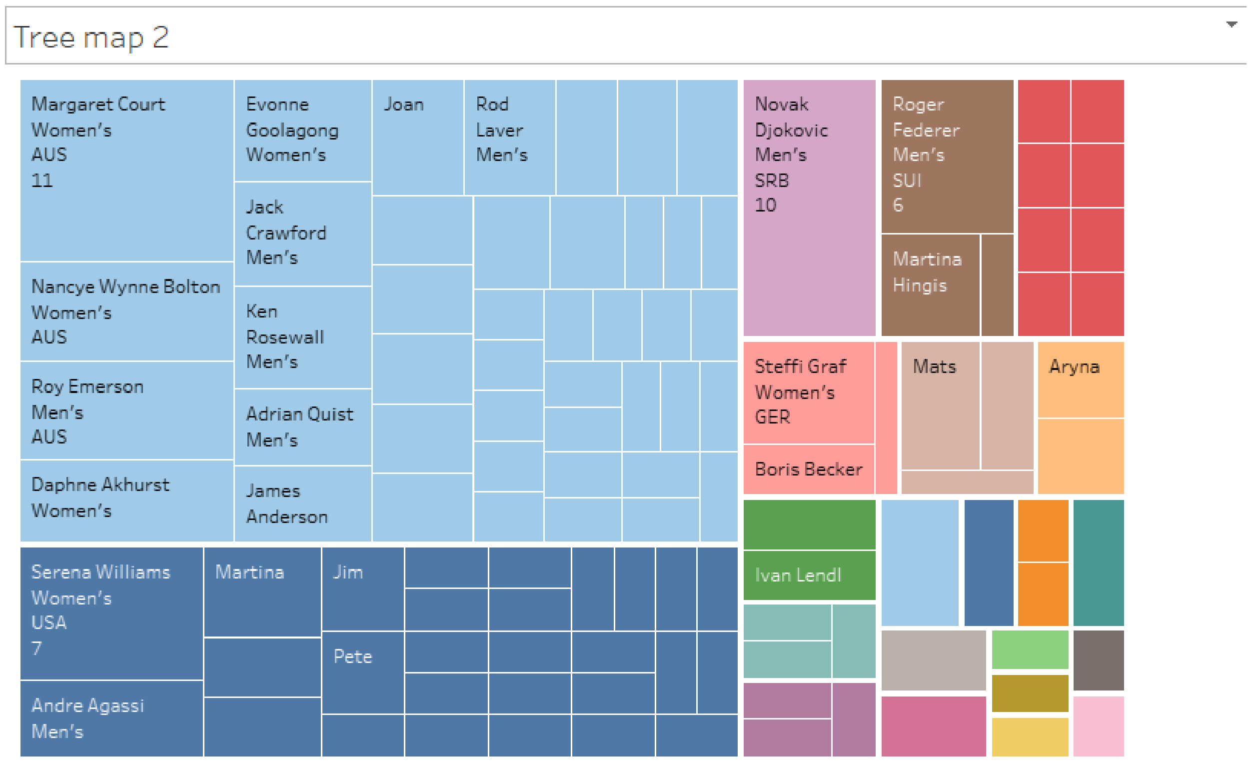
The Tree Map effectively visualizes the relationship between gender, champion nationality, and win rate for Australian Open champions, using nested rectangles where the size represents the number of championships won and colors distinguish between nationalities. This allows for a clear comparison of top-performing players and nations, with Australia dominating in terms of championships, followed by the United States, and Novak Djokovic standing out with the most wins. The visualization is space-efficient, enabling quick comparisons of wins across countries and genders, and color-coding enhances clarity in identifying national trends. However, while the Tree Map is excellent for visualizing large



values, it can be challenging to compare smaller values accurately, and it doesn't always convey exact numbers as effectively as other chart types. Despite these limitations, the Tree Map is a powerful tool for revealing broader patterns and hierarchies within the dataset, providing valuable insights into player and country performance at the Australian Open.

Figure- 2 (Champion Nationality, Gender and 1st won

The Tree Map 2 in your Tableau file builds upon the initial visualization by focusing more specifically on the Champion Nationality, gender, and the 1st set won data. In this visualization, each rectangle represents a player's performance based on the number of 1st sets they've won in Australian Open finals, while the size of the rectangles reflects the number of championships won. The colors differentiate between the nationalities of the champions, making it easier to identify the dominance of specific countries.One of the key insights from Tree Map 2 is the clear indication of Australia's dominance in men's finals, as the largest rectangles correspond to Australian male champions. The visualization shows that both male and female Australian players have a significant presence, with the highest number of 1st set wins coming

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from Australia, followed by the United States. This pattern continues for women, where Australia holds a substantial share in the 1st set victories, demonstrating that their players not only win championships but also frequently secure the crucial first sets.Tree Map 2 is useful for understanding the importance of winning the first set in a championship match, as it often sets the tone for the rest of the match. It also visually highlights how countries like Australia and the United States consistently produce champions who excel in these key moments. However, as with the previous tree map, this visualization's limitations include difficulties in comparing smaller values, as rectangles representing players with fewer wins are harder to distinguish. Nonetheless, Tree Map 2 provides a valuable snapshot of first-set performance by country and gender, emphasizing the early dominance of top champions in Australian Open finals.

Advantages of Tree Map:

1. **Efficient Space Usage**: Tree maps allow a large amount of information to be visualized in a compact space, making them ideal for visualizing complex, high-dimensional data without overwhelming the viewer.
2. **Hierarchical Visualization**: They provide a clear way to represent hierarchical relationships within data, making it easy to see parent-child relationships and how different categories are related.
3. **Quick Comparisons**: Tree maps make it easy to compare the relative size of different categories or subcategories. The size of the rectangles corresponds to data values, providing a visual way to assess their relative significance.
4. **Customizable**: Tree maps can be tailored to specific datasets through the use of color schemes, sizes, and labels, enhancing their ability to communicate complex patterns.
5. **Good for Overview**: They offer a quick overview of large datasets, helping users spot trends, patterns, or outliers at a glance.

Disadvantages of Tree Map:

1. **Limited Precision**: While useful for visual comparisons, tree maps are not as precise as other visualization techniques, such as bar charts or line graphs, when conveying exact numerical values.
2. **Difficulty with Small Values**: Small data values may not be represented effectively, as the rectangles corresponding to smaller categories can be too small to distinguish or compare accurately.
3. **Less Intuitive for Some Users**: For users unfamiliar with hierarchical visualizations, tree maps may be less intuitive and require additional time to understand the relationships between rectangles.
4. **Clutter with Too Much Data**: When dealing with very large datasets or too many categories, tree maps can become cluttered, making it hard to distinguish between smaller categories or identify meaningful insights.

5. **Size Doesn't Convey Full Story**: The size of a rectangle only represents a single data point (e.g., the number of wins), but it may not communicate more complex relationships or distributions within the data.

2 Geographic Map

The Geographic Map provides a powerful visualization tool for analyzing data based on location, such as champion nationality in the Australian Open dataset. Geographic maps are effective for visually representing how the distribution of championships is spread across different countries. In your files, each country that has produced an Australian Open champion is marked on the map, with circles or markers used to show the number of wins by players from that country. The size of the markers typically reflects the number of championships, with larger markers indicating more wins.

In this visualization, **Longitude** and **Latitude** serve as the coordinates to place the countries on the map. The map is color-coded to differentiate countries, making it easier to visually interpret which nations have had the most success in the Australian Open. For example, Australia and the United States have the largest markers, indicating their dominance in the tournament, as they have produced the most champions over the years.By interacting with the map, users can gain insights into how different countries have performed over time. For instance, a user can hover over the markers to view details about the champions from each country, their gender, and their win rates.

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The visualization helps reveal patterns such as Australia's consistent success, with many top champions emerging from both its male and female divisions, or the growing representation of champions from other countries like Switzerland, Russia, and Spain.

Advantages:

Geographical Context: The Geographic Map provides a clear, intuitive way to see how championships are distributed globally. It helps to visualize the concentration of tennis success in specific regions, such as Australia, North America, and Europe.

Easy Interpretation: Geographic maps are familiar to most users, making them accessible even to people who are not experts in data visualization. The use of size and color coding allows for quick visual interpretation of data distribution across different countries.

Insightful Comparison: The map allows for easy comparison of countries' performance, showing which regions have been more successful and how players from different nationalities contribute to the tournament's history.

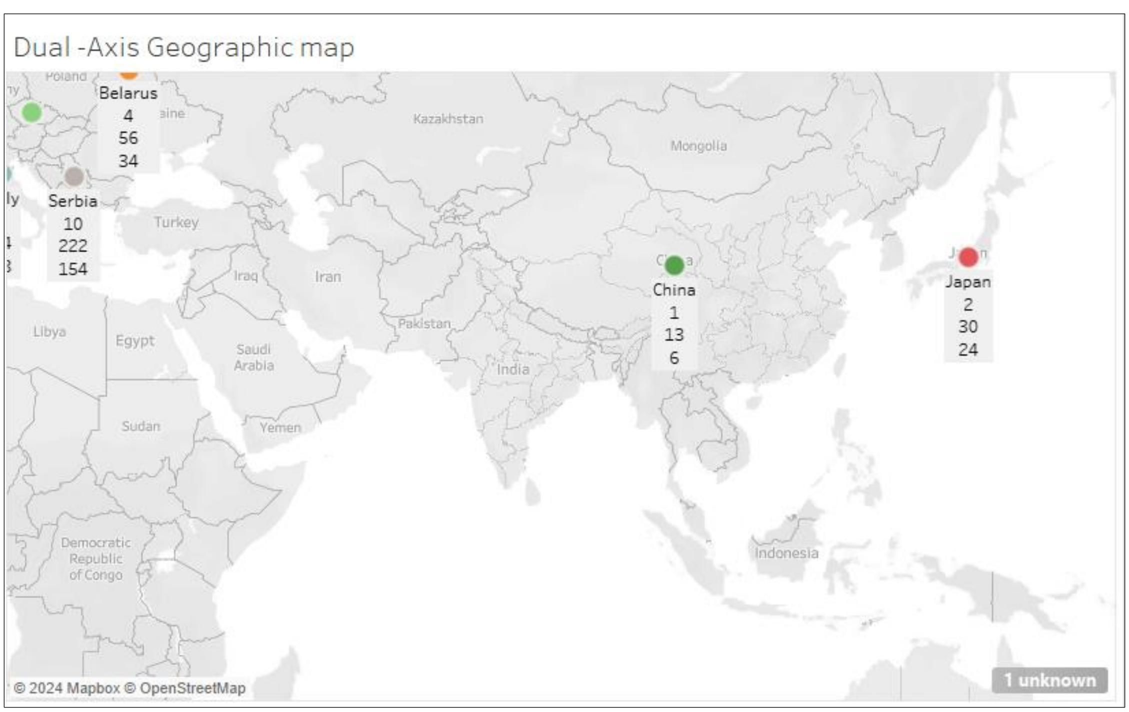
Disadvantages:

Limited to Geographic Data: The map is most effective for data that has a clear geographic component, which can limit its use for other types of analysis where location is not a factor.

Overemphasis on Location: The visual impact of the map may overemphasize the importance of geographic location, potentially drawing focus away from other key performance factors like individual player statistics.

Clutter with Multiple Data Points: If many countries have small numbers of wins, their markers may overlap or become difficult to distinguish, which could make the map crowded and less informative in those regions.

3 Dual Axis Geographic Map



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The Dual Axis Geographic Map provides a compelling method for visualizing two different sets of data simultaneously on a single map. In the context of the Australian Open dataset, this visualization is used to compare and analyze two geographical variables, such as champion nationality and win rate, or gender and performance metrics across different countries. By leveraging a dual-axis setup, this map combines both datasets onto the same geographic space, enhancing the ability to spot patterns and relationships between the two variables. Dual Axis Geographic Map consists of two overlapping maps, each representing a different set of data. For example, one axis might plot the nationality of champions with markers or circles sized according to the number of wins from each country. The second axis can overlay additional data, such as the win rate or gender, using different markers or color schemes to distinguish between the variables. This dual visualization allows for direct comparisons between the datasets. For instance, by mapping champion nationality alongside win rates, you can quickly see how certain countries not only produce a high number of champions but also tend to have higher win rates, highlighting their dominance in the tournament. Similarly, plotting gender alongside the same geographic data can reveal patterns in how male and female champions are distributed across the world

Advantages:

Comparison of Two Variables: The dual-axis map enables the visualization of two different datasets on the same geographic space, making it easy to compare and analyze their relationships.

Enhanced Data Depth: By overlaying two different sets of information, the map provides a more comprehensive view of the data. For example, you can see not just where champions come from but also how well they perform.

Clear Visual Patterns: Geographic maps are intuitive and familiar, and the addition of dual axes makes it easier to spot correlations, such as how countries with more champions also tend to have higher win rates or how gender is distributed across regions.

Disadvantages:

Complexity: While the dual-axis map adds more depth to the data, it can also increase the complexity of the visualization. It may be harder for users to immediately interpret the relationships between the two datasets, especially if the markers or color schemes overlap or become cluttered.

Overemphasis on Geography: As with all geographic maps, the dual-axis map might overemphasize location as the main factor in the analysis, potentially distracting from other important performance metrics or variables.

Data Overlap: With two sets of data displayed on the same map, there is a risk of overlap or clutter, particularly if the data points are closely related or if the markers are not clearly differentiated.

4 Scatter chart

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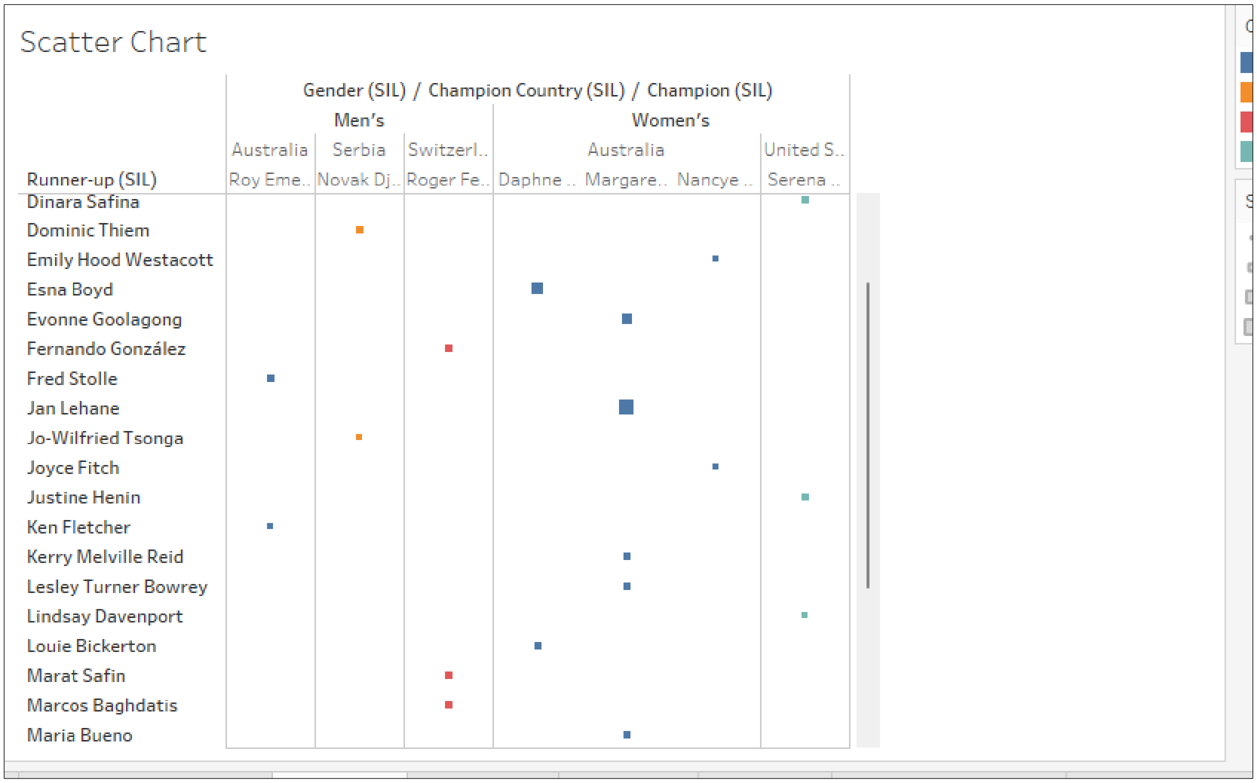

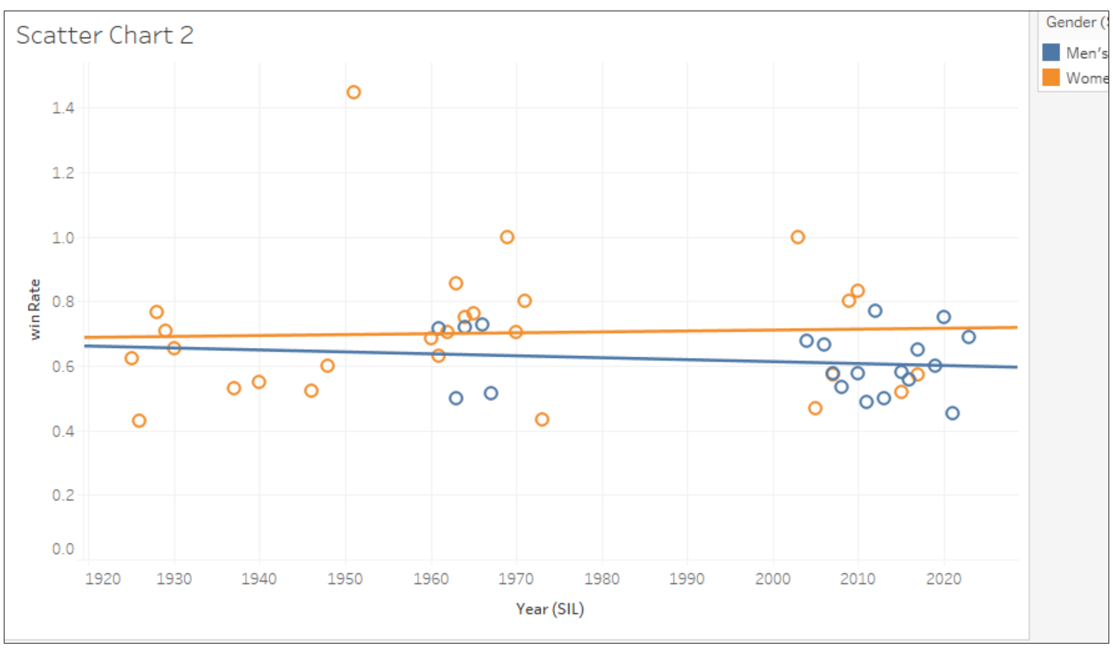

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Scatter Chart 1 is used to visually explore the relationship between two key variables, such as win rate and sets won, across different Australian Open champions. Each point on the chart represents a player, with the placement determined by the values of these two metrics. This chart helps reveal

patterns, such as whether winning more sets consistently leads to a higher win rate, and highlights outliers like players who perform exceptionally well or poorly in comparison to others.

Scatter Chart 2 builds on this by adding a third variable, such as match duration or nationality, using color and marker size to differentiate players or countries. For example, players from the same country may be grouped by color, providing insight into whether certain nationalities dominate in specific areas. This multidimensional approach allows for a more comprehensive analysis, revealing how different factors (like long matches or a player's nationality) impact performance. Both scatter charts are valuable for identifying trends, comparing players, and understanding the relationships between multiple variables within the dataset.

Advantages of Scatter Chart and Scatter Chart 2:

Reveals Relationships: Both charts effectively show correlations between variables like win rate and sets won.

Identifies Outliers: They easily highlight outliers and exceptional performances.

Clear and Simple: Scatter Chart 1 is straightforward, while Scatter Chart 2 adds depth with a third variable (e.g., nationality) using color and marker size.

Multidimensional Analysis: Scatter Chart 2 allows richer analysis by adding an extra dimension for deeper insights.

Disadvantages of Scatter Chart and Scatter Chart 2:

Limited Dimensions: Both are limited in showing more than two or three variables and can become cluttered with more data.

Overlapping Points: Large datasets can cause overlap, making it harder to interpret.

No Grouping: They don't inherently group or cluster data, requiring additional tools.

Shows Correlation, Not Causality: They show relationships but don't prove cause and effect

5 Lollopop chart

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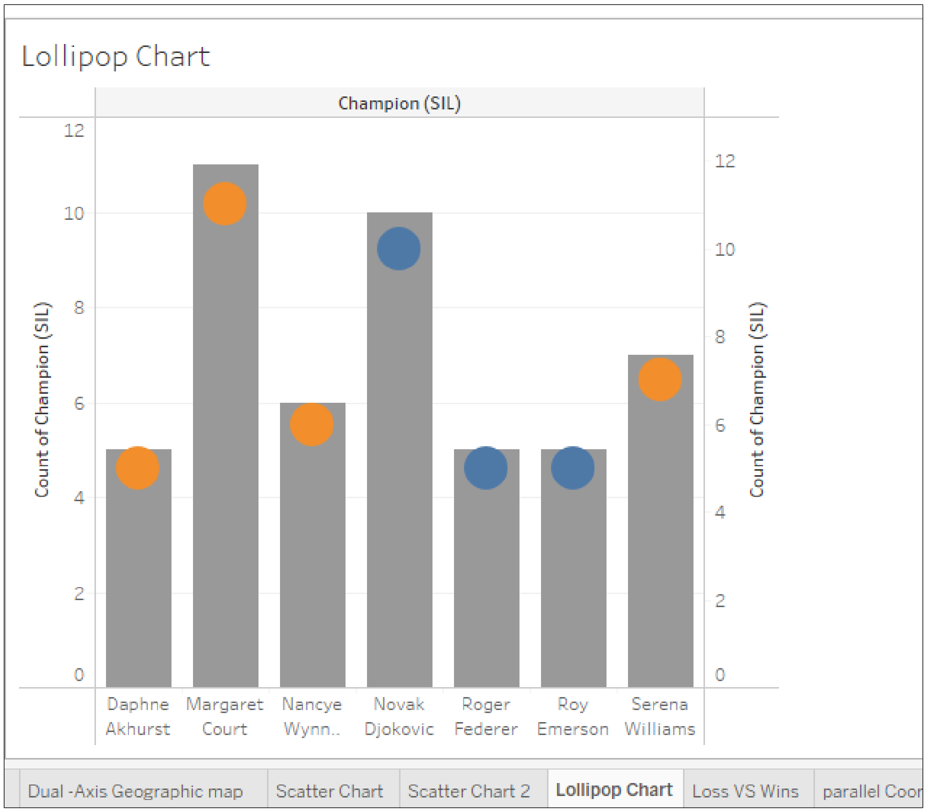

The **Lollipop Chart** is a variation of a bar chart that uses a line with a circle (or "lollipop") to represent data points, making it ideal for comparing categories like **win rates** or **sets won** in the Australian Open dataset. Categories are displayed along the horizontal axis, with the line showing the value and the circle marking the data point. This chart is visually appealing, easy to interpret, and helps highlight comparisons between players or nationalities.

Advantages:

Visual Clarity: It's cleaner and less cluttered than a bar chart, offering clear comparisons.

Customizable: Colors and circle sizes can encode additional dimensions, like gender or nationality.

Disadvantages:

Limited Precision: It may not convey exact values as well as bar charts.

Less Effective for Large Data: Too many categories can make it cluttered and harder to interpret.

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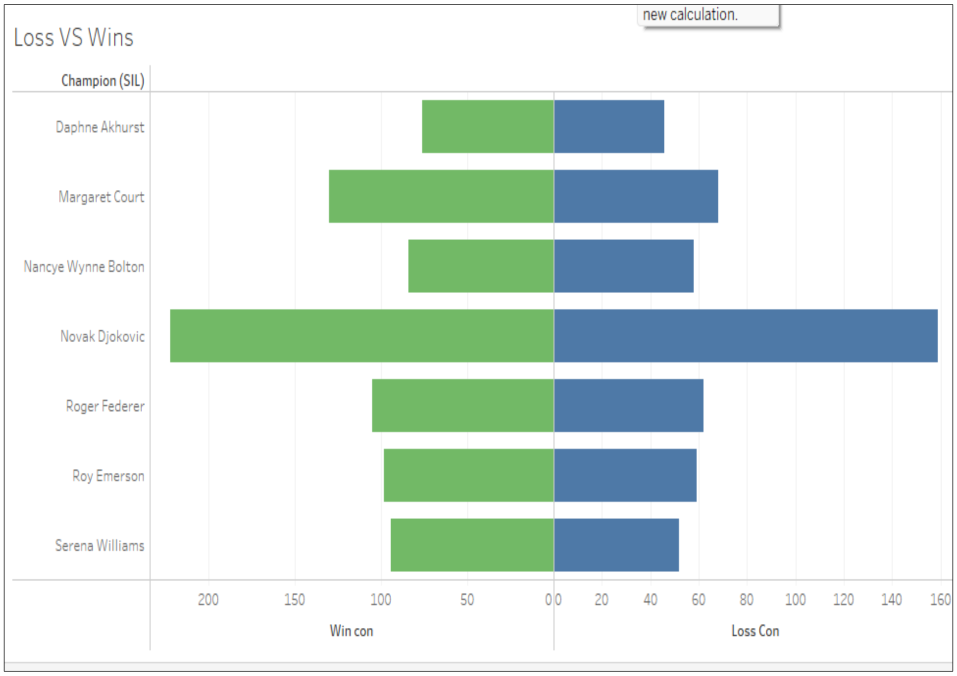


6 Loss VS Win

The **Loss and Win Chart** in your Tableau file visually compares the **total wins** and **losses** of players in the Australian Open dataset. Each player's performance is represented by two bars or markers—one for wins and one for losses—making it easy to see the balance between victories and defeats. This chart allows for quick identification of top performers with more wins and highlights players with higher loss rates.

Advantages:

Clear Comparison: Easily compares wins and losses for each player.

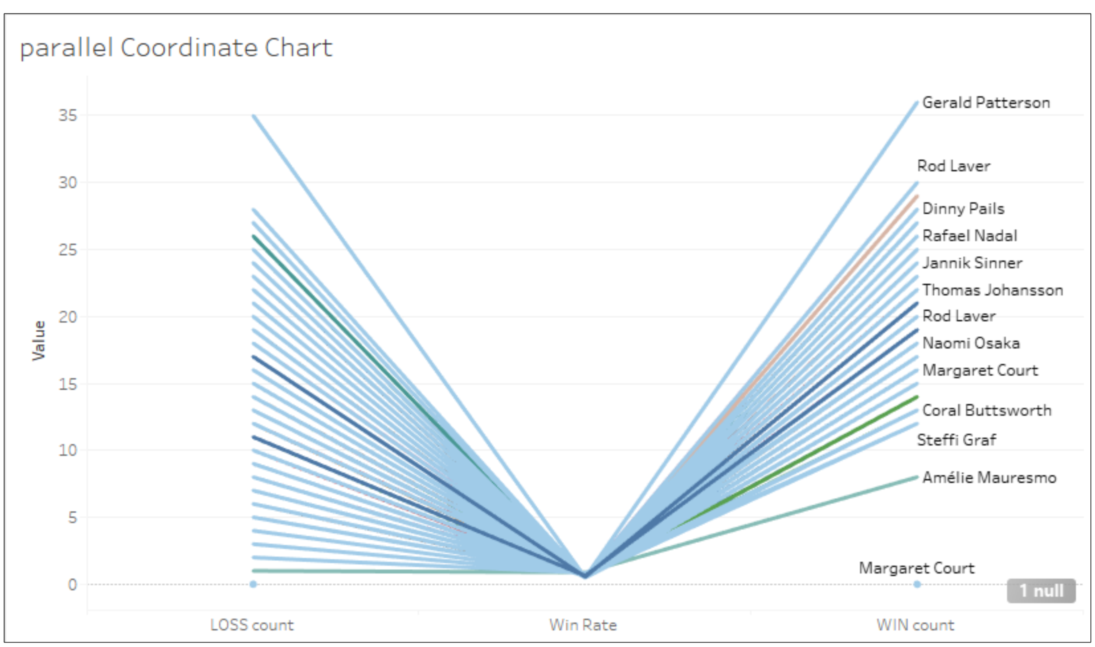
Insightful: Helps quickly identify top players and those with more losses.

Disadvantages:

Limited Detail: Focuses on win-loss ratio without showing other performance metrics.

Cluttered for Large Data: With many players, it can become crowded and harder to interpret.

7 Parallel coordinate chart



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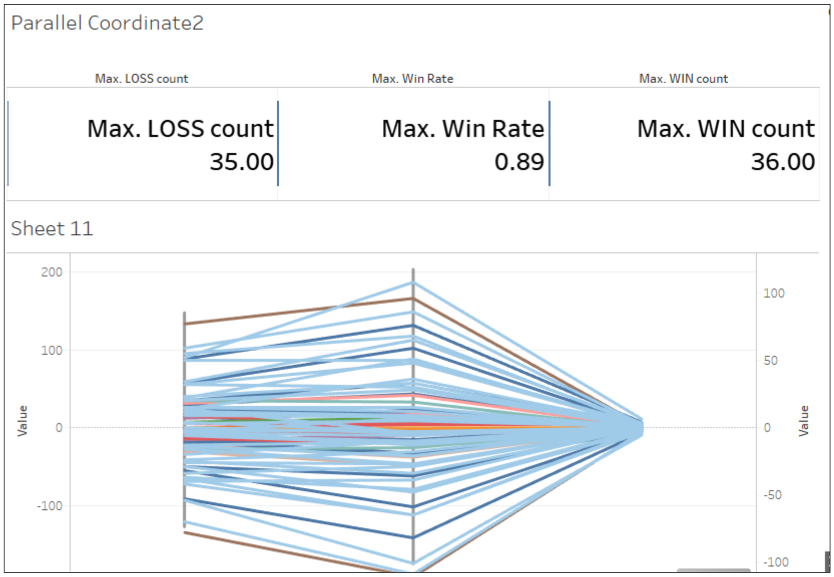

The **Parallel Coordinate** chart is used to visualize the relationships between multiple variables, such as **sets won**, **win rate**, and **match duration**, for Australian Open players. Each variable is represented by a vertical axis, and lines connect individual data points across these axes, allowing you to identify trends and correlations between multiple metrics. This type of chart is particularly useful for high-dimensional data, helping to uncover patterns that might not be obvious when looking at single variables.

The **Parallel Coordinate 2** chart adds more depth by incorporating additional variables, such as **nationality** or **gender**, often through color coding or line thickness. This enhanced version allows for a more nuanced analysis, making it easier to differentiate between different groups or categories in the dataset while still comparing multiple variables simultaneously.

Advantages:

Multidimensional Analysis: Easily compare several variables at once, spotting trends, correlations, and outliers.

Customization: Parallel Coordinate 2 adds clarity by incorporating dimensions like nationality through colors and line thickness.

Interactive: Users can highlight or filter specific data points, aiding targeted analysis.

Disadvantages:

Clutter: Adding more data can cause overlapping lines, making interpretation difficult.

Scaling Issues: Poor axis scaling can distort relationships.

Less Intuitive: Requires more understanding compared to simpler charts and may overwhelm new

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Conclusion

This report presents an advanced exploration of the Australian Open dataset, employing a range of sophisticated data visualization techniques to analyze key performance indicators such as **win rates**, **sets won**, **match durations**, and the influence of **player nationality** and **gender**. Through the use of visual tools like the **Tree Map**, **Parallel Coordinate Charts**, **Scatter Plots**, and **Geographic Maps**, we were able to uncover complex patterns, such as the long-standing dominance of players from Australia and the United States, and examine how factors such as nationality and gender correlate with performance metrics over time. These visualizations not only revealed global patterns of success but also helped isolate top-performing individuals and outliers, offering deeper insight into championship trends.

The **Lollipop Chart** and **Win/Loss Chart** were particularly effective in illustrating comparative performance metrics, providing a clear visual representation of success rates across players, while maintaining a high degree of clarity and simplicity. In contrast, the **Parallel Coordinate Charts** enabled a more nuanced and multidimensional analysis of multiple variables simultaneously, allowing for the identification of correlations and trends that would otherwise be difficult to detect. However, as with any complex visualization, there is an inherent challenge in balancing the richness of the data with the clarity of the presentation, especially when handling larger datasets prone to visual clutter.

By integrating these diverse visualization techniques, the report provides a holistic and multi-faceted understanding of the Australian Open dataset, highlighting the value of data-driven insights in sports analytics. The ability to compare and contrast player performance across multiple dimensions—such as nationality, gender, and historical trends—affords a richer, more detailed comprehension of how champions emerge and sustain success over time. Ultimately, this analysis demonstrates how data visualization can transform raw, high-dimensional data into actionable insights, enhancing decision-making and uncovering underlying patterns that are otherwise difficult to perceive. This approach not only underscores the strategic importance of visual analytics in sports performance evaluation but also serves as a model for applying these techniques to similarly complex datasets across other domains.

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